

Ch 5. Probability

- Basic Concepts – Review of Set Theory and Venn Diagram
- Counting – Ordered, Unordered, Sampling With and Without Replacement.
- Probability Theory
- Base Rate: Bayes' Theorem

- Set Theory - Review

A set is a collection of elements.

Eg:

$A = \{\text{Head}, \text{Tail}\}, \text{Head} \in A, \text{Tail} \in A$

$B = \{1, 2, 3, 4, 5, 6\}$

$C = \{1, 3, 5\}$

- Set Theory - Review

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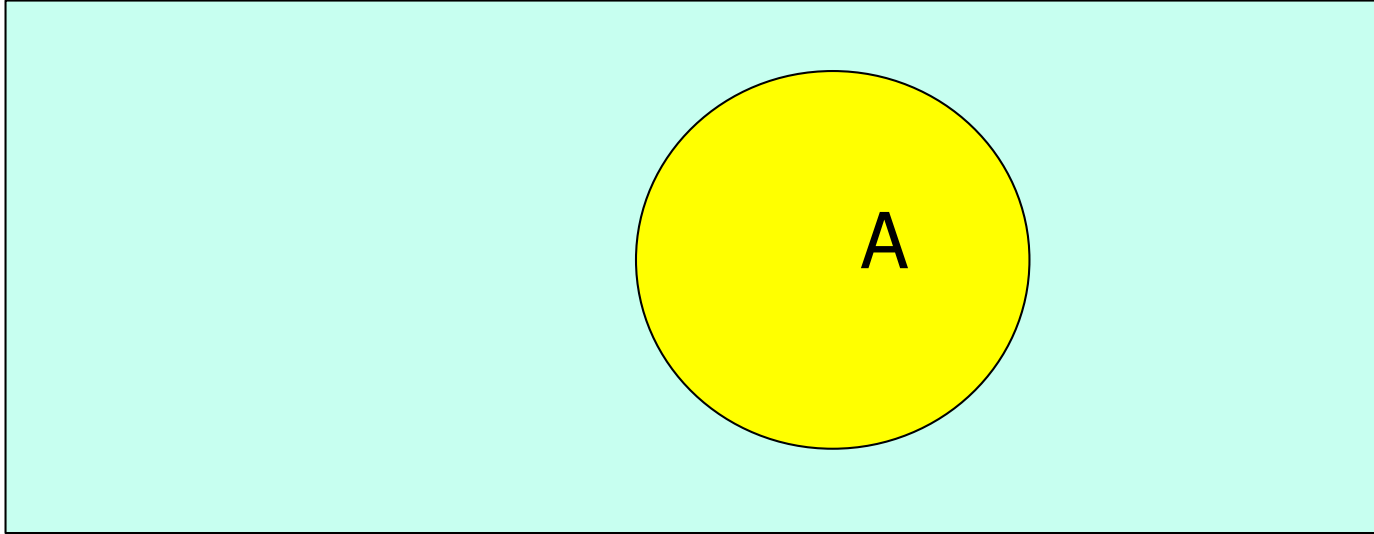
Set C is a subset of Set B since all elements in C are also in B. Notation: $C \subset B$

The set of all elements in a given context is known as a universal set Ω .

Empty or null set ϕ is a set with no element.

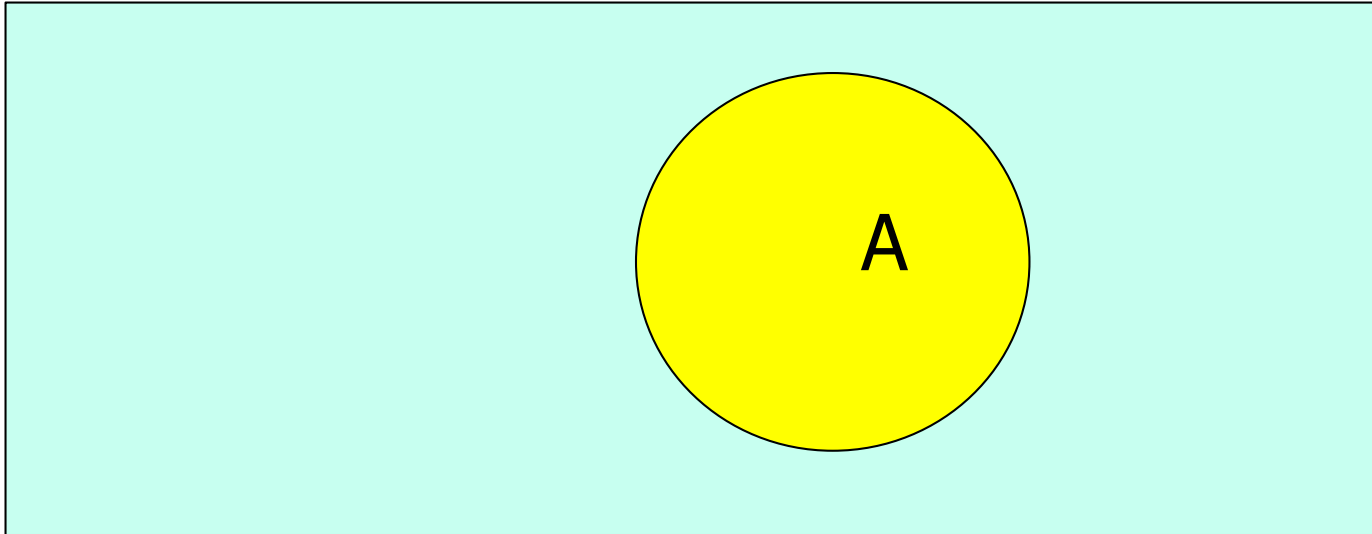
- Venn Diagram

universal set Ω



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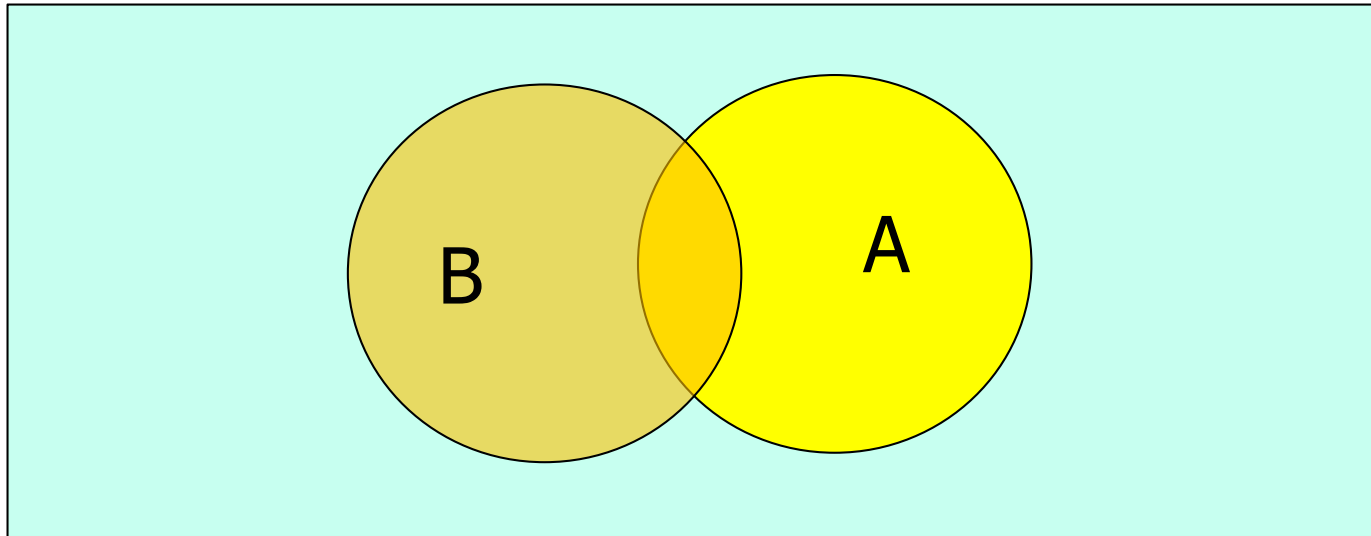


Set $A \subset \Omega$. Complement of A , denoted by A' , consists of all elements in Ω that are not in A .

Eg: $\Omega = \{1,2,3,4,5,6\}$, $A = \{1,2\}$, $A' = \{3,4,5,6\}$

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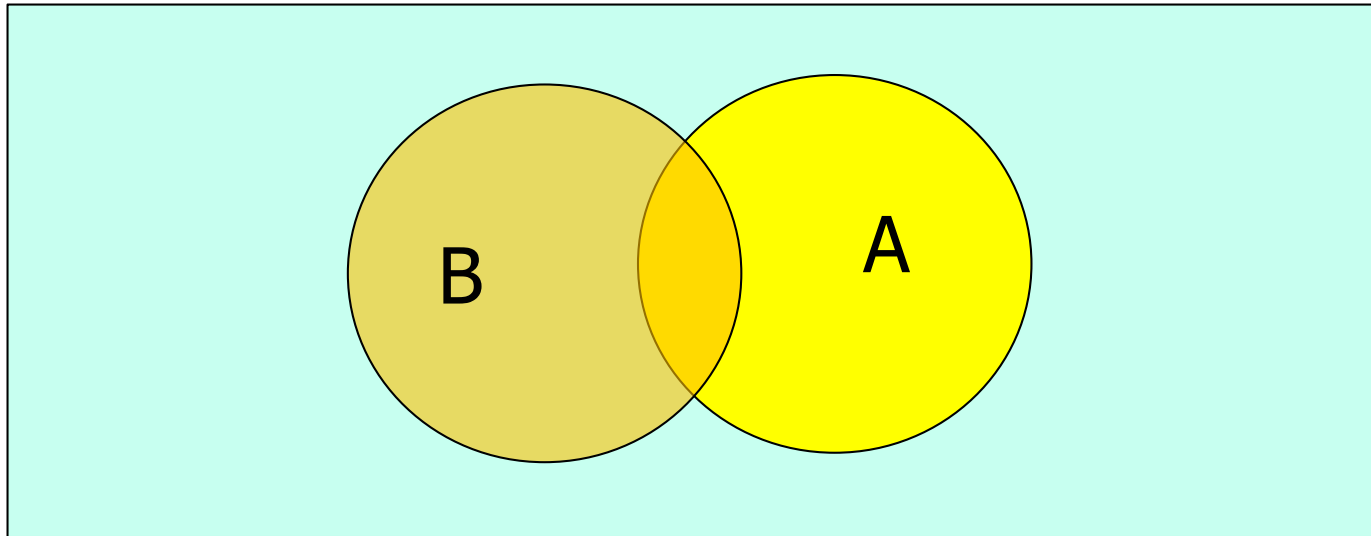
Eg: $\Omega = \{1,2,3,4,5,6\}$, $A = \{1,2\}$, $A' = \{3,4,5,6\}$

Union of A and B consists of all elements in A or B. Notation: $A \cup B$.

Eg: $\{1,2,3\} \cup \{2,3,4\} = \{1,2,3,4\}$

■ Venn Diagram

universal set Ω



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Intersection of A and B consists of all elements in both A and B. Notation: $A \cap B$.

Eg: $\{1,2,3\} \cap \{2,3,4\} = \{2,3\}$

Consider rolling a die. Let the outcome set $A = \{1, 2, 3\}$ and set $B = \{2, 4, 6\}$. Match the following:

Go to **wooclap.com** and use the code **QTINLE**

Consider rolling a die. Let the outcome set $A = \{1, 2, 3\}$ an...

$A \cup B =$	1	A	$\{1, 2, 3, 4, 6\}$
$A \cap B =$	2	B	$\{4, 6\}$
$A' \cap B =$	3	C	$\{2\}$
$A' \cap B' =$	4	D	$\{5\}$

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- Counting

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Different scenarios for consideration:

- sampling with replacement
- sampling without replacement
- digits drawn are ordered
- digits drawn are unordered

- Counting – general scenario

Sampling

Eg:

Pick 3 numbers
(one at a time)
from a set of 10
digits.

Arrangement can
be ordered or
unordered

Two options:
with or without replacement.

After picking each number
from the set, the number can
be replaced or no
replacement.

- Counting –
 1. Ordered sampling with replacement

Ordered sampling of k elements from a set of n elements with replacement.

No. of possible arrangements = n^k

- Counting –

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No. of possible arrangements = n^k

Eg: Calculate the total number of different 4-digit numeric codes. Each digit is independently selected from $\{0,1,2,\dots,9\}$.

1st digit: 10 possible numbers

2nd digit: also 10 possible numbers.

3rd and 4th digit: each also 10 possible numbers.

Total arrangements = $10 \times 10 \times 10 \times 10 = 10^4$

- Counting –
 2. Ordered sampling without replacement

Ordered sampling of k elements from a set of n elements without replacement.

No. of possible arrangements =

$$n \times (n - 1) \times (n - 2) \times (n - 3) \times \dots \times (n - k + 1)$$

Number of k permutations of n elements:

$${}_nP_k = \frac{n!}{(n-k)!} = \frac{n \times (n-1) \times \dots \times 1}{(n-k) \times (n-k-1) \dots \times 1}$$

Eg: Calculate the total number of different 4-digit numeric codes. Each digit is sampled from $\{0,1,2,\dots,9\}$ without replacement.

1st digit: 10 possible numbers

2nd digit: 9 possible numbers

3rd digit: 8 possible numbers and so on.

Total arrangements = $10 \times 9 \times 8 \times 7$

i.e. there are ${}^{10}P_4 = \frac{n!}{(n-k)!} = \frac{10!}{(10-4)!}$ arrangements

Eg: 4 boys and 2 girls sit in a row. Find the following:

Go to **wooclap.com** and use the code **QTINLE** 

4 boys and 2 girls sit in a row. Find the following: 

(1) No. of ways of putting these 6 people in a row. 

(2) No. of ways such that each girl has a boy to her left and to her right. 




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- Counting –
 3. Unordered sampling without replacement

We choose k elements from a set of n elements and the ordering does not matter.

Number of arrangements equals to k combinations of n elements ${}^nC_k = \binom{n}{k} = \frac{n!}{\textcolor{red}{k!} (n-k)!}$

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Eg: Using digits from a set $\{1, 2, 3, 4, 5, 6\}$, how many combinations of two numbers can we make?

Ans: $\binom{6}{2} = \frac{6!}{2! (6-2)!} = \frac{6 \times 5}{2 \times 1} = 15$

- Counting –
 4. Unordered sampling with replacement

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Eg: Pick two numbers from $\{1, 2, 3\}$:

Possible outcomes: $(1,1)$, $(1,2)$, $(1,3)$, $(2,2)$, $(2,3)$ and $(3,3)$

We have 6 combinations with $n=3$ and $k=2$.

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We have 6 combinations with $n=3$ and $k=2$.

In general: No. of combinations = $\binom{n + k - 1}{k}$

Eg: A 6-bit code is made up of 4 1's and 2 0's.
Calculate:

Go to **wooclap.com** and use the code **QTINLE** 

A 6-bit code is made up of four '1's and two '0's. Find the ... 

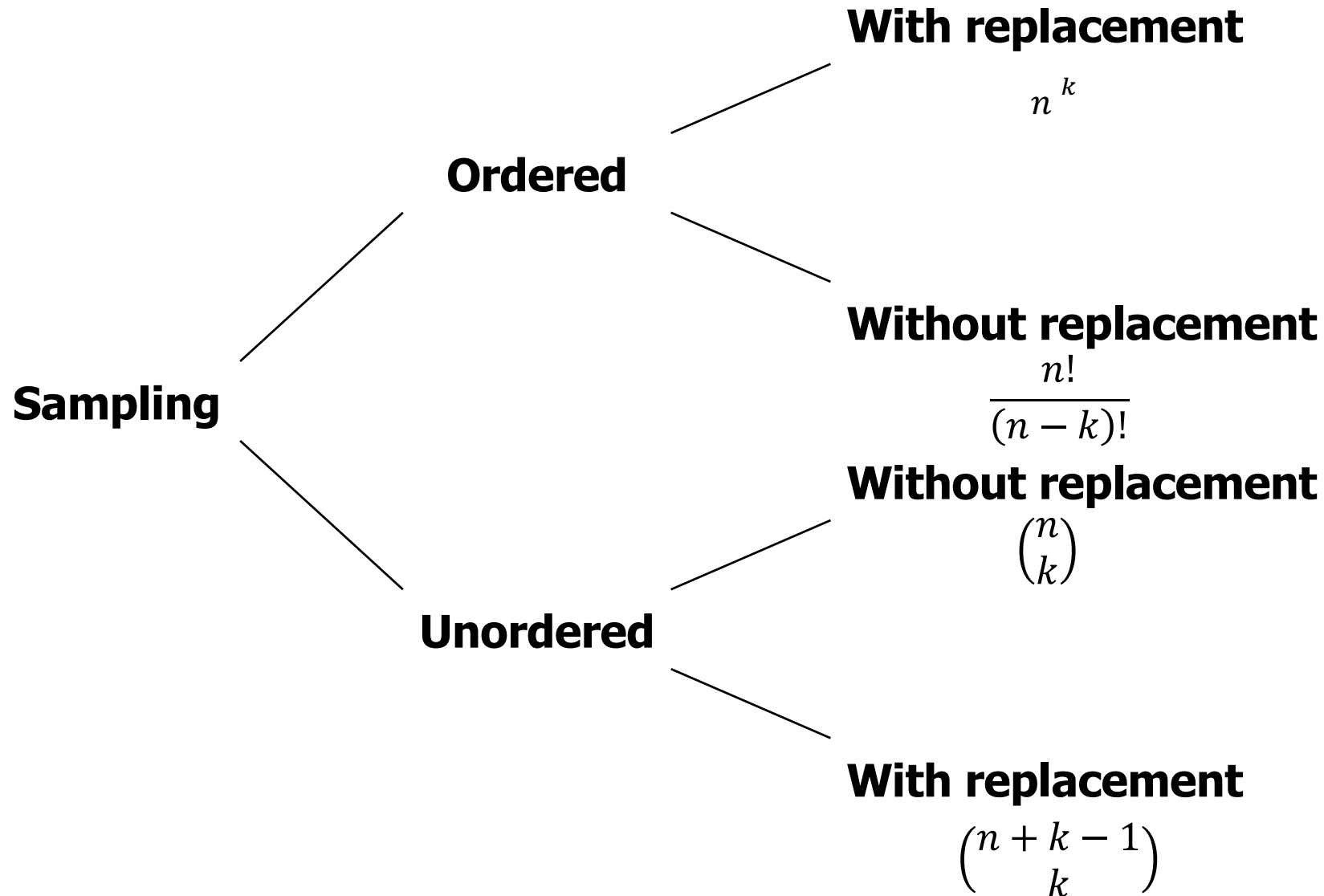
(1) No. of distinct codewords in the code. 

(2) No. of codewords such that a '0' has a '1' to its left and to its right. 

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- Counting –
No. of arrangements/combinations



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Total of $6 \times 6 = 36$ possible outcomes.

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Total of $6 \times 6 = 36$ possible outcomes.

Probability of getting a "1" followed by another "1" is $1/36$.

Probability of getting 1st no. = 2nd no. is $6/36$.

Definitions

- A sample space is the set S of all possible outcomes of an experiment.
- An event is a set of one or more (favorable) outcomes in the sample space.

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$\text{Event } X = \{1, 3, 5\}$$

$$\text{Event } Y = \{3, 6\}$$

Definitions

- A sample space is the set S of all possible outcomes of an experiment.
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- Two events are mutually exclusive if they have no outcomes in common.

Event $X = \{1, 3, 5\}$

Event $Y = \{3, 6\}$

are not mutually exclusive.

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If two events A and B are mutually exclusive,
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If two events A and B are independent,
$$P(A \cap B) = P(A)P(B).$$

E.g. Probability of getting a "1" followed by another "1" is $1/36 = 1/6 \times 1/6$.

Axioms of Probability:

- (1) Probability of any outcome or event X is non-negative:

$$P(X) \geq 0$$

- (2) Probability of the sample space S is 1:

$$P(S) = 1$$

- (3) If $X_1, X_2, X_3, \dots$ are mutually exclusive events, then:

$$P(X_1 \text{ or } X_2 \text{ or } X_3 \dots) = P(X_1) + P(X_2) + P(X_3) + \dots$$

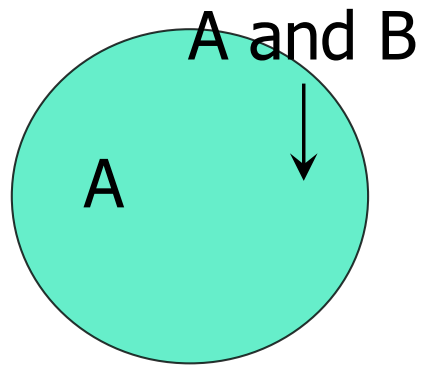
Mutually exclusive events and probabilities

Eg: A company has decided that in the next 5 years, 40% of their new employees will be men, 30% will be Singaporean, and 35% will be foreigner women. What is the probability that a new employee will be a Singaporean man?

	Men	Women	Total
Foreigner		0.35	
Singaporean	$0.4 - 0.35$ $= 0.05$	$0.6 - 0.35$ $= 0.25$	0.3
Total	0.4	$1 - 0.4$ $= 0.6$	

Non-mutually exclusive events and probabilities

Eg: A certain kind of fruit is grown in 2 districts, A and B. Both areas sometimes get fruitflies. Suppose the probabilities are $P(A)=0.1$, $P(B)=0.05$ and $P(A \text{ and } B)=0.02$, what is the probability that one or other (or both) districts are infected at a given time?



$$\begin{aligned} P(A \text{ or } B) &= P(A) + P(B) - P(A \text{ and } B) \\ &= 0.1 + 0.05 - 0.02 \\ &= 0.13 \end{aligned}$$

Probabilities and Events (independent)

Eg: A company has 2 guards. Each carries a pager activated by sensors. Guard 1 and guard 2 respond to pager alert 80% and 50% of the time respectively. They independently report any alert. What is the probability that at least one will report an alert?

Mutually exclusive and independent events

Q: Is it possible to have mutually exclusive events which are independent?

Ans: No, if A and B are mutually exclusive events, knowing A occurs implies B cannot occur.

Worked Examples considered so far:

Probabilities & Events (mutually exclusive)

Probabilities & Events (non-mutually exclusive)

Probabilities & Events (independent)

Next: Consider Conditional Probabilities

i.e. Probability of an event A given that an event B has occurred, denoted as $P(A|B)$

Conditional Probabilities

Assuming that 10 percent of the days are rainy in a certain city.

$$P(\text{rain}) = 0.1$$

The probability that it rains given that it is cloudy might be, say,

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$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \text{ where } P(B) > 0$$

Conditional Probabilities

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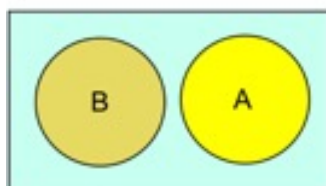


Fig 1.

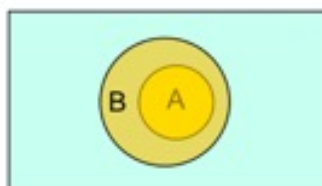


Fig 2.



Given that $P(A) = 0.2$ and $P(B) = 0.5$, determine the following.

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Conditional Probabilities

Eg: A jar contains 10 blue, 5 red, 4 green and 1 yellow marbles. Two marbles are randomly picked. What is the probability that one will be blue and the other yellow?

Bayes' Theorem

Given two events A and B, where $P(A) > 0$:

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

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E.g.

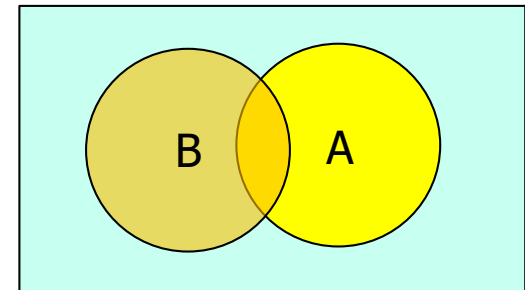
$$P(spam|words) = \frac{P(words|spam)P(spam)}{P(words)}$$

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Note: $P(A) = P(A \cap B) + P(A \cap B')$

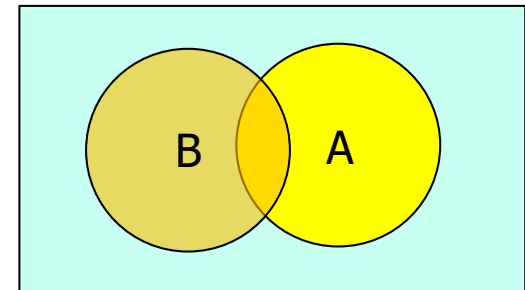


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Similarly:

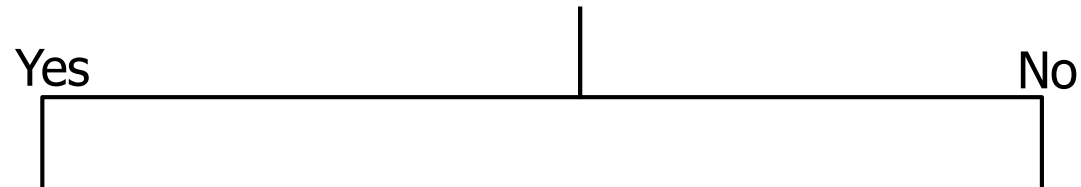
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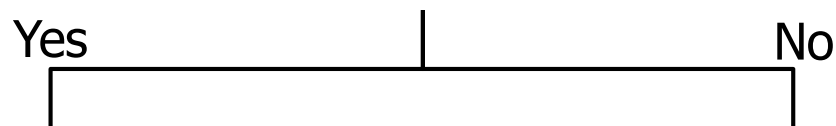
Eg: A test correctly identifies a disease in 95% of people who have it. It correctly identifies no disease in 94% of people who do not have it. In the population, 3% of the people have the disease. What is the probability that one has the disease if tested positive?

Summary: Probability Calculations

Are X and Y mutually exclusive?

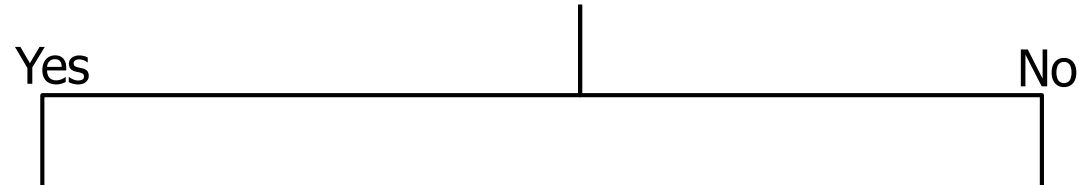


Are X and Y independent?



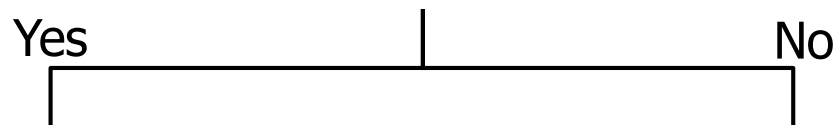
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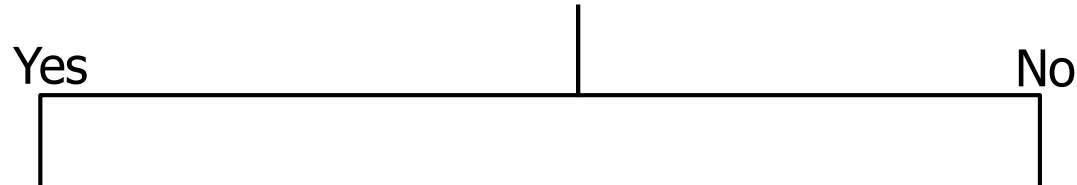
1. $P(X \text{ and } Y) = 0$ because event (X and Y) is impossible
2. $P(X \text{ or } Y) = P(X) + P(Y)$

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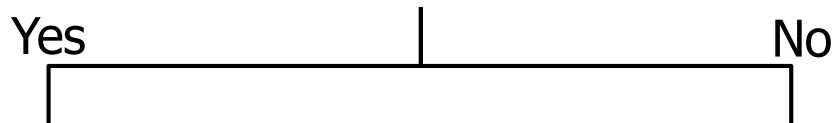
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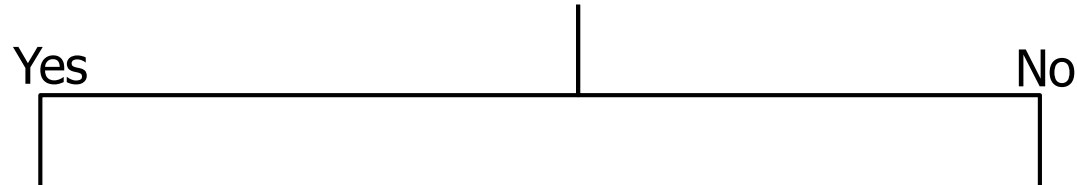
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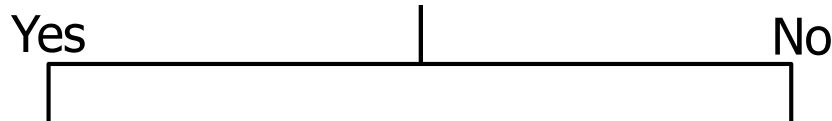
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- Yes
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- No
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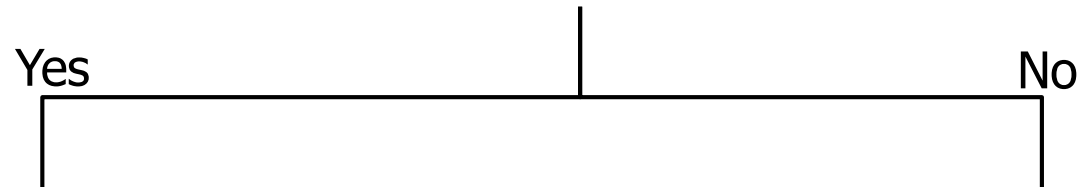
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 2. $P(X|Y) = P(X)$

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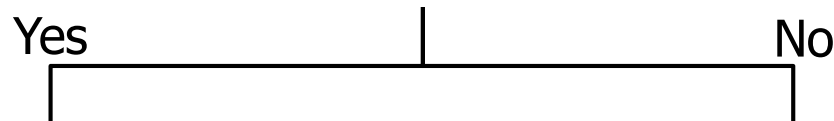
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2. $P(X \text{ and } Y) = P(X|Y) P(Y)$
 $= P(Y|X) P(X)$